INCH-POUND
MIL-DTL-24643/33F
1 October 2009
SUPERSEDING
MIL-DTL-24643/33E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS2SWU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight

First	-	Copper conductor, tin-coated, 18 AWG, Class B stranding (7 by 0.0152 inches) in accordance with ASTM B8.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white and one black conductor for each pair.
Fourth	-	Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2.
Eighth	-	The specified number of pairs (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Ninth	-	An optional binder.
Tenth	-	Cross-linked polyolefin jacket (see table I for thickness).

NOTE: For LS2SWU-1, omit seventh and eighth.

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TABLE I. Details.

Military part no. M24643/33	Type and size	Number of pairs	Cable jacket thickness min avg. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UN	LS2SWU-1	1	0.015	0.240	0.255
-02UN	LS2SWU-3	3	0.075	0.670	0.710
-03UN	LS2SWU-7	7	0.075	0.860	0.910
-04UN	LS2SWU-12	12	0.088	1.130	1.200
-05UN	LS2SWU-19	19	0.100	1.292	1.380
-06UN	LS2SWU-24	24	0.100	1.500	1.590
-07UN	LS2SWU-30	30	0.100	1.670	1.760
-08UN	LS2SWU-37	37	0.125	1.785	1.870
-09UN	LS2SWU-61	61	0.125	2.205	2.300

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	7.47
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Component shield to component shield	500
Insulation resistance (megohms.1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure
Hydrostatic (open end) (leakage at 300 lb/in^2, 2 hours, in^3, max.)	1.0
Capacitance	
Mutual capacitance, at 1 kilohertz (KHz) (pF/ft, max.)	30
Capacitance unbalance (percent max.)	8

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Characteristic impedance, at 1 megahertz (MHz) (ohms)	75±5
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75

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Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LS2SWU	1 through 12	1500
LS2SW	19 through 24	1000
LS2SWU	30 through 61	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-037)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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